

Bengaluru North University

Solution of Question Bank

I B.Sc Mathematics-I(SEP)

Unit-II: Differential Calculus-I

2 Marks Questions:

1. Find the nth derivative of the following

i) $y = 7^{3x}$

$$y_n = 3^n 7^{3x} (\log 7)^n$$

ii) $y = e^{-5x}$

$$y_n = (-5)^n e^{-5x}$$

iii) $y = \log(1 - 2x)$

$$y_n = \frac{(-1)^n (n-1)! (-2)^n}{(1-2x)^n} = \frac{(n-1)! 2^n}{(1-2x)^n}$$

iv) $y = \frac{1}{3x-1}$

$$y_n = \frac{(-1)^n n! 3^n}{(3x-1)^{n+1}}$$

v) $y = \cos(1 - 4x)$

$$y_n = (-4)^n \cos\left(1 - 4x + \frac{n\pi}{2}\right)$$

vi) $y = \frac{1}{(5x-2)^3}$

where $a = 5, b = -2$ and $p = 3$

$$\begin{aligned}\therefore y_n &= \frac{(-1)^n (3+n-1)!}{(3-1)!} 5^n (5x-2)^{-(3+n)} \\ &= \frac{(-1)^n (n+2)!}{2!} 5^n \frac{1}{(5x-2)^{n+3}}\end{aligned}$$

vii) $y = e^{-2x} \sin(6x + 3)$

This is of the form $e^{ax} \sin(bx + c)$ where $a = -2, b = 6, c = 3$

$$\begin{aligned}y_n &= ((-2)^2 + 6^2)^{\frac{n}{2}} e^{-2x} \sin\left(6x + 3 + n \tan^{-1}\left(\frac{6}{-2}\right)\right) \\ &= 40^{\frac{n}{2}} e^{-2x} \sin(6x + 3 - n \tan^{-1} 3)\end{aligned}$$

viii) $y = \sin 3x \cos 2x$

$$y = \frac{1}{2} (2 \sin 3x \cos 2x)$$

$$y = \frac{1}{2} (\sin 5x + \sin x)$$

$$\therefore y_n = \frac{1}{2} \left[5^n \sin \left(5x + \frac{1}{2} n\pi \right) + 1^n \sin \left(x + \frac{1}{2} n\pi \right) \right]$$

$$\text{ix) } y = \log \sqrt{\frac{2x+1}{x-2}}$$

$$y = \frac{1}{2} [\log(2x+1) - \log(x-2)]$$

$$y_n = \frac{1}{2} [(-1)^{n-1} (n-1)! 2^n (2x+1)^{-n} - (-1)^{n-1} (n-1)! 1^n (x-2)^{-n}]$$

$$\text{x) } y = \log(x^2 - 4)$$

$$y = \log(x+2) + \log(x-2)$$

$$y = \log(x+2) + \log(x-2)$$

$$y_n = [(-1)^{n-1} (n-1)! 1^n (x+2)^{-n} + (-1)^{n-1} (n-1)! 1^n (x-2)^{-n}]$$

2. Find the nth derivative of $y = e^x \log x$

Solution : By Leibnitz theorem, we have

$$\begin{aligned} (e^x \log x)_n &= (e^x)_n \log x + {}^n C_1 (e^x)_{n-1} \frac{1}{x} + {}^n C_2 (e^x)_{n-2} \left(-\frac{1}{x^2} \right) \\ &\quad + \dots + e^x (-1)^{n-1} (n-1)! x^{-n} \\ &= e^x \log x + {}^n C_1 e^x \cdot \frac{1}{x} - {}^n C_2 e^x \cdot \frac{1}{x^2} + {}^n C_3 2! x^{-3} - \dots + e^x (-1)^{n-1} (n-1)! x^{-n} \\ &= e^x \left[\log x + \frac{n}{x} - \frac{n(n-1)}{2x^2} + \dots + \frac{(-1)^{n-1} (n-1)!}{x^n} \right] \end{aligned}$$

3. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for the following functions.

i) $z = x^3 + y^3 + 3x^2y$

Differentiating partially w.r.t x , $\frac{\partial z}{\partial x} = 3x^2 + 6xy$

Differentiating partially w.r.t y , $\frac{\partial z}{\partial y} = 3y^2 + 3x^2$

ii) $z = \log(x^2 + y^2)$

Differentiating partially w.r.t x , $\frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2}$

Differentiating partially w.r.t y , $\frac{\partial z}{\partial y} = \frac{2y}{x^2 + y^2}$

iii) $z = e^{x/y}$

Differentiating partially w.r.t x , $\frac{\partial z}{\partial x} = e^{x/y} \left(\frac{1}{y} \right)$

Differentiating partially w.r.t y , $\frac{\partial z}{\partial y} = e^{x/y} \left(\frac{-x}{y^2} \right)$

iv) $z = x \cos(xy) + y \sin(xy)$

Differentiating partially w.r.t x , $\frac{\partial z}{\partial x} = x(-\sin(xy) y) + \cos(xy) \cdot 1 + y \cos(xy) \cdot y$
 $= -xy \sin(xy) + \cos(xy) + y^2 \cos(xy)$

Differentiating partially w.r.t y , $\frac{\partial z}{\partial y} = x(-\sin(xy) x) + \sin(xy) + y \cos(xy) \cdot y$
 $= -x^2 \sin(xy) + \sin(xy) + y^2 \cos(xy)$

v) $z = \tan^{-1} \left(\frac{y}{x} \right)$

Differentiating partially w.r.t x , $\frac{\partial z}{\partial x} = \frac{1}{1 + \frac{y^2}{x^2}} \left(\frac{-y}{x^2} \right) = \frac{x^2}{x^2 + y^2} \left(\frac{-y}{x^2} \right) = \frac{-y}{x^2 + y^2}$

Differentiating partially w.r.t y , $\frac{\partial z}{\partial y} = \frac{1}{1 + \frac{y^2}{x^2}} \left(\frac{1}{x} \right) = \frac{x^2}{x^2 + y^2} \left(\frac{1}{x} \right) = \frac{x}{x^2 + y^2}$

vi) $z = e^{xy}$

Differentiating partially w.r.t x , $\frac{\partial z}{\partial x} = e^{xy} y = ye^{xy}$

Differentiating partially w.r.t y , $\frac{\partial z}{\partial y} = e^{xy} x = xe^{xy}$

4. If $u = \frac{x^3 + y^3}{\sqrt{x^2 + y^2}}$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$

$$u = \frac{x^3 + y^3}{\sqrt{x^2 + y^2}} = \frac{x^3 \left(1 + \frac{y^3}{x^3} \right)}{x \sqrt{1 + \frac{y^2}{x^2}}} = x^2 \phi \left(\frac{y}{x} \right)$$

$\therefore u$ is a homogeneous function of degree 2. By Euler's theorem $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$

5. If $u = \sin^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x}$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$

Solution: Given $u = \sin^{-1} \left(\frac{x}{y} \right) + \tan^{-1} \left(\frac{y}{x} \right) = \operatorname{cosec}^{-1} \left(\frac{y}{x} \right) + \tan^{-1} \left(\frac{y}{x} \right)$
 $= \text{a function of } \left(\frac{y}{x} \right) = \phi \left(\frac{y}{x} \right)$
 $= x^0 \phi \left(\frac{y}{x} \right)$
 $\Rightarrow u$ is a homogeneous function of degree 0 in x and y .
 By Euler's theorem, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0 \cdot u = 0$.

6. IF $u = x^3 + y^3 + z^3 - 3xyz$, then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3u$

$$\begin{aligned} u(\lambda x, \lambda y, \lambda z) &= (\lambda x)^3 + (\lambda y)^3 + (\lambda z)^3 - 3\lambda x \lambda y \lambda z \\ &= \lambda^3(x^3 + y^3 + z^3 - 3xyz) \\ &= \lambda^3 u \end{aligned}$$

$\Rightarrow u$ is a homogeneous function of degree 3 in x, y, z .

By Euler's theorem, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 3u$

7. Find the total differential of $z = x^3y + x^2y^2 + xy^3$

Solution: Given $f(x, y) = x^3y + x^2y^2 + xy^3$

We have $\frac{\partial f}{\partial x} = 3x^2y + 2xy^2 + y^3$; $\frac{\partial f}{\partial y} = x^3 + 2x^2y + 3xy^2$

Then $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = (3x^2y + 2xy^2 + y^3)dx + (x^3 + 2x^2y + 3xy^2)dy$

8. Find the total differential of $z = xsiny - ysinx$.

We have $\frac{\partial z}{\partial x} = siny - ycosx$ and $\frac{\partial z}{\partial y} = xcosy - sinx$

Then $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = (siny - ycosx)dx + (xcosy - sinx)dy$

9. Find the total differential of $f(x, y) = \tan^{-1} \frac{y}{x}$

Given $f(x, y) = \tan^{-1} \frac{y}{x}$

We have $\frac{\partial f}{\partial x} = \frac{1}{1 + (\frac{y}{x})^2} \left(\frac{-y}{x^2} \right) = \frac{-y}{x^2 + y^2}$ and $\frac{\partial f}{\partial y} = \frac{1}{1 + (\frac{y}{x})^2} \left(\frac{1}{x} \right) = \frac{x}{x^2 + y^2}$

Then $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy = \frac{xdy - ydx}{x^2 + y^2}$

10. Find the total differential of $f(x, y) = x + y + z + xyz$

Given $f(x, y) = \tan^{-1} \frac{y}{x}$

We have $\frac{\partial f}{\partial x} = 1 + yz$ and $\frac{\partial f}{\partial y} = 1 + xz$, $\frac{\partial f}{\partial z} = 1 + xy$

Then $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = (1 + yz)dx + (1 + xz)dy + (1 + xy)dz$

11. Find $\frac{dy}{dx}$ using partial differentiation given $\sin(x + y) + \cos(x - y) = y$.

Let $F(x, y) = \sin(x + y) + \cos(x - y) - y$.

so that $F(x, y) = 0$

$F_x = \cos(x + y) - \sin(x - y)$

$F_y = \cos(x + y) + \sin(x - y) - 1$

$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{(\cos(x + y) - \sin(x - y))}{(\cos(x + y) + \sin(x - y) - 1)}$

12. Find $\frac{dy}{dx}$ using partial differentiation given $x^3 + y^3 - 3axy = 0$.

Let $F(x, y) = x^3 + y^3 - 3axy$ so that $F(x, y) = 0$

$F_x = 3x^2 - 3ay$

$F_y = 3y^2 - 3ax$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{(x^2 - ay)}{(y^2 - ax)}$$

13. Find $\frac{dy}{dx}$ using partial differentiation given $e^x \sin y + e^y \sin x = 1$

Let $F(x, y) = e^x \sin y + e^y \sin x - 1$ so that $F(x, y) = 0$

$$F_x = e^x \sin y + e^y \cos x$$

$$F_y = e^x \cos y + e^y \sin x$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{(e^x \sin y + e^y \cos x)}{(e^x \cos y + e^y \sin x)}$$

14. If $u = 3x + 5y, v = 4x - 3y$ show that $\frac{\partial(u,v)}{\partial(x,y)} = -29$

$$\text{Given } u = 3x + 5y \Rightarrow \frac{\partial u}{\partial x} = 3, \frac{\partial u}{\partial y} = 5$$

$$v = 4x - 3y \Rightarrow \frac{\partial v}{\partial x} = 4, \frac{\partial v}{\partial y} = -3$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 3 & 5 \\ 4 & -3 \end{vmatrix} = 3 \times -3 - 5 \times 4 = -9 - 20 = -29$$

15. If $u = x + \frac{y^2}{x}, v = \frac{y^2}{x}$ then find $\frac{\partial(u,v)}{\partial(x,y)}$.

$$\text{Given } u = x + \frac{y^2}{x} \Rightarrow \frac{\partial u}{\partial x} = 1 - \frac{y^2}{x^2}, \frac{\partial u}{\partial y} = \frac{2y}{x}$$

$$v = \frac{y^2}{x} \Rightarrow \frac{\partial v}{\partial x} = -\frac{y^2}{x^2}, \frac{\partial v}{\partial y} = \frac{2y}{x}$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 - \frac{y^2}{x^2} & \frac{2y}{x} \\ -\frac{y^2}{x^2} & \frac{2y}{x} \end{vmatrix} = 1 - \frac{y^2}{x^2} \times \frac{2y}{x} - \frac{2y}{x} \times -\frac{y^2}{x^2} = \frac{2y}{x} - \frac{2y^3}{x^3} + \frac{2y^3}{x^3} = \frac{2y}{x}$$

16. If $x = r \cos \theta, y = r \sin \theta$ show that $\frac{\partial(x,y)}{\partial(r,\theta)} = r$

$$\text{Given } x = r \cos \theta \Rightarrow \frac{\partial x}{\partial \theta} = -r \sin \theta, \frac{\partial x}{\partial r} = \cos \theta$$

$$y = r \sin \theta \Rightarrow \frac{\partial y}{\partial \theta} = r \cos \theta, \frac{\partial y}{\partial r} = \sin \theta$$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

17. If $u = x^2 - 2y, v = x + y$ show that $\frac{\partial(u,v)}{\partial(x,y)} = 2(x + 1)$

$$\text{Given } u = x^2 - 2y \Rightarrow \frac{\partial u}{\partial x} = 2x, \frac{\partial u}{\partial y} = -2$$

$$v = x + y \Rightarrow \frac{\partial v}{\partial x} = 1, \frac{\partial v}{\partial y} = 1$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & -2 \\ 1 & 1 \end{vmatrix} = 2x + 2 = 2(x + 1)$$

5 Marks Questions

1. Find the nth derivative of $y = e^{ax} \sin(bx + c)$

$$\text{Let } y = e^{ax} \sin(bx + c)$$

$$\begin{aligned}\Rightarrow y_1 &= e^{ax} b \cos(bx + c) + \sin(bx + c) a e^{ax} \\ &= e^{ax} [b \cos(bx + c) + a \sin(bx + c)]\end{aligned}$$

$$\text{Put } a = r \cos \theta \text{ and } b = r \sin \theta \text{ then } r^2 = a^2 + b^2, r = \sqrt{a^2 + b^2} \text{ and}$$

$$\theta = \tan^{-1} \frac{b}{a}$$

$$\begin{aligned}y_1 &= e^{ax} [r \sin \theta \cos(bx + c) + r \cos \theta \sin(bx + c)] \\ &= r e^{ax} [\sin(bx + c) \cos \theta + \cos(bx + c) \sin \theta] \\ &= r e^{ax} \sin(bx + c + \theta)\end{aligned}$$

$$\begin{aligned}y_2 &= r [e^{ax} b \cos(bx + c + \theta) + \sin(bx + c + \theta) a e^{ax}] \\ &= r e^{ax} [r \sin \theta \cos(bx + c + \theta) + r \cos \theta \sin(bx + c + \theta)] \\ &= r^2 e^{ax} [\sin(bx + c + \theta) \cos \theta + \cos(bx + c + \theta) \sin \theta] \\ &= r^2 e^{ax} \sin(bx + c + 2\theta)\end{aligned}$$

Continuing this way, we get

$$y_n = r^n e^{ax} \sin(bx + c + n\theta)$$

$$\therefore y_n = (a^2 + b^2)^{n/2} e^{ax} \sin\left(bx + c + n \tan^{-1} \frac{b}{a}\right)$$

2. Find the nth derivative of $y = e^{ax} \cos(bx + c)$

$$\text{Let } y = e^{ax} \cos(bx + c)$$

$$\begin{aligned}\Rightarrow y_1 &= -e^{ax} b \sin(bx + c) + \cos(bx + c) a e^{ax} \\ &= e^{ax} [-b \sin(bx + c) + a \cos(bx + c)]\end{aligned}$$

$$\text{Put } a = r \cos \theta \text{ and } b = r \sin \theta \text{ then } r^2 = a^2 + b^2, r = \sqrt{a^2 + b^2} \text{ and}$$

$$\theta = \tan^{-1} \frac{b}{a}$$

$$\begin{aligned}y_1 &= e^{ax} [-r \sin \theta \sin(bx + c) + r \cos \theta \cos(bx + c)] \\ &= r e^{ax} [\cos(bx + c) \cos \theta - \sin(bx + c) \sin \theta] \\ &= r e^{ax} \cos(bx + c + \theta)\end{aligned}$$

$$\begin{aligned}y_2 &= r [-e^{ax} b \sin(bx + c + \theta) + \cos(bx + c + \theta) a e^{ax}] \\ &= r e^{ax} [r \cos \theta \cos(bx + c + \theta) - r \sin \theta \sin(bx + c + \theta)]\end{aligned}$$

$$= r^2 e^{ax} [\cos(bx + c + \theta) \cos \theta - \sin(bx + c + \theta) \sin \theta]$$

$$= r^2 e^{ax} \cos(bx + c + 2\theta)$$

Continuing this way, we get

$$y_n = r^n e^{ax} \cos(bx + c + n\theta)$$

$$\therefore y_n = (a^2 + b^2)^{n/2} e^{ax} \cos\left(bx + c + n \tan^{-1} \frac{b}{a}\right)$$

3. Find the nth derivative of $y = \cos x \cos 2x \cos 3x$

$$\text{Let } y = \cos x \cos 2x \cos 3x$$

$$= \frac{1}{2} \cos x (2 \cos 2x \cos 3x)$$

$$= \frac{1}{2} \cos x (\cos 5x + \cos x)$$

$$= \frac{1}{4} (2 \cos x \cos 5x + 2 \cos^2 x)$$

$$y = \frac{1}{4} (\cos 6x + \cos 4x + \cos 2x + 1)$$

$$\therefore y_n = \frac{1}{4} \left[6^n \cos\left(6x + \frac{n\pi}{2}\right) \right] + 4^n \cos\left(4x + \frac{n\pi}{2}\right) + 2^n \cos\left(2x + \frac{n\pi}{2}\right)$$

4. Find the nth derivative of $y = \sin x \sin 2x \sin 3x$

$$\text{Let } y = \sin x \sin 2x \sin 3x$$

$$= \frac{1}{2} \sin x (2 \sin 2x \sin 3x)$$

$$= -\frac{1}{2} \sin x (\cos 5x - \cos x)$$

$$= -\frac{1}{4} (2 \sin x \cos 5x - 2 \sin x \cos x)$$

$$y = -\frac{1}{4} (\sin 6x - \sin 4x - \sin 2x)$$

$$y_n = -\frac{1}{4} \left[6^n \sin\left(6x + \frac{n\pi}{2}\right) \right] - 4^n \sin\left(4x + \frac{n\pi}{2}\right) - 2^n \sin\left(2x + \frac{n\pi}{2}\right)$$

5. Find the nth derivative of $y = e^x \sin x \cos 2x$

$$\text{Let } y = e^x \sin x \cos 2x$$

$$= \frac{e^x}{2} (2 \sin x \cos 2x)$$

$$= \frac{e^x}{2} (\sin 3x - \sin x)$$

$$= \frac{1}{2} [e^x \sin 3x - e^x \sin x]$$

$$y_n = \frac{1}{2} \left[10^{n/2} e^x \sin\left(3x + n \tan^{-1} \frac{3}{1}\right) - 2^{n/2} e^x \sin\left(x + n \frac{\pi}{4}\right) \right]$$

6. Find the nth derivative of $y = \sinh 2x \cdot \sin 4x$

$$\text{Let } y = \sinh 2x \cdot \sin 4x$$

$$= \left(\frac{e^{2x} - e^{-2x}}{2} \right) \sin 4x$$

$$= \frac{1}{2} [e^{2x} \sin 4x - e^{-2x} \sin 4x]$$

$$y_n = \frac{1}{2} [20^{n/2} e^{2x} \sin(4x + n \tan^{-1} 2) - 20^{n/2} e^{-2x} \sin(4x - n \tan^{-1} 2)]$$

7. Find the nth derivative of $y = \frac{1}{(x+2)(x-1)}$

iii) Let $y = \frac{1}{(x+2)(x-1)}$
 resolving the given fraction into partial functions

$$\frac{1}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1} \Rightarrow 1 = A(x-1) + B(x+2)$$

 put $x = 1 \Rightarrow B = 1/3$
 put $x = -2 \Rightarrow A = -1/3$

$$\therefore \frac{1}{(x+2)(x-1)} = -\frac{1/3}{x+2} + \frac{1/3}{x-1}$$

$$y = -\frac{1/3}{x+2} + \frac{1/3}{x-1} = \frac{1}{3} \left[-\frac{1}{x+2} + \frac{1}{x-1} \right]$$

$$\left(\text{For } y = \frac{1}{ax+b} \text{ we have } y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}} \right)$$

Hence $y_n = \frac{1}{3} \left[-\frac{(-1)^n n!}{(x+2)^{n+1}} + \frac{(-1)^n n!}{(x-1)^{n+1}} \right]$

8. Find the nth derivative of $y = \frac{1}{6x^2 - 5x + 1}$

iv) Let $y = \frac{1}{6x^2 - 5x + 1}$

$$6x^2 - 5x + 1 = 6x^2 - 3x - 2x + 1 = 3x(2x-1) - 1(2x-1) = (3x-1)(2x-1)$$

$$\frac{1}{(3x-1)(2x-1)} = \frac{A}{3x-1} + \frac{B}{2x-1} \Rightarrow 1 = A(2x-1) + B(3x-1)$$

 put $x = \frac{1}{2} \Rightarrow B = 2$, put $x = \frac{1}{3} \Rightarrow A = -3$

$$y = \frac{1}{6x^2 - 5x + 1} = \frac{A}{3x-1} + \frac{B}{2x-1} = \frac{-3}{3x-1} + \frac{2}{2x-1}$$

$$y_n = -\frac{3(-1)^n n! 3^n}{(3x-1)^{n+1}} + \frac{2(-1)^n n! 2^n}{(2x-1)^{n+1}}$$

$$y_n = (-1)^n n! \left(\frac{2^{n+1}}{(2x-1)^{n+1}} - \frac{3^{n+1}}{(3x-1)^{n+1}} \right)$$

9. Find the nth derivative of $y = \frac{1}{(x-1)^2(x-2)}$

6) Let $y = \frac{x^2}{(x-1)^2(x-2)} = \frac{-3}{x-1} - \frac{1}{(x-1)^2} + \frac{4}{(x-2)}$ (after resolving to partial fraction)

$$\therefore y_n = \frac{(-3)(-1)^n n!}{(x-1)^{n+1}} - \frac{(-1)^n (n+1)!}{(x-1)^{n+2}} + \frac{4(-1)^n n!}{(x-2)^{n+1}}$$

10. Find the nth derivative of $y = \frac{1}{1+3x+2x^2}$

1) Let $y = \frac{x}{1+3x+2x^2} = \frac{x}{(1+x)(1+2x)}$ resolving into partial fractions,

we get $y = \frac{1}{(x+1)} - \frac{1}{(2x+1)}$. Using the standard result $\frac{d^n}{dx^n} \left(\frac{1}{ax+b} \right)$ we get,

$$y_n = \frac{(-1)^n n!}{(x+1)^{n+1}} - \frac{(-1)^n n! 2^n}{(2x+1)^{n+1}} = (-1)^n n! \left[\frac{1}{(x+1)^{n+1}} - \frac{2^n}{(2x+1)^{n+1}} \right]$$

11. Find the nth derivative of $y = \frac{1}{(x+2)(x-2)^2}$

Let $y = \frac{x-1}{(x-2)^2(x+2)} = \frac{3}{16} \cdot \frac{1}{(x-2)} + \frac{1}{4} \cdot \frac{1}{(x-2)^2} - \frac{3}{16} \cdot \frac{1}{(x+2)}$

(after resolving into partial fraction)

$$\therefore y_n = \frac{3}{16} \frac{(-1)^n n!}{(x-2)^{n+1}} + \frac{1}{4} \frac{(-1)^n (n+1)!}{(x-2)^{n+2}} - \frac{3}{16} \frac{(-1)^n n!}{(x+2)^{n+1}}$$

12. State and prove Leibnitz's Theorem

2.4 Leibnitz's Theorem:

Statement : If $y = uv$ where u and v are functions of x having n^{th} order derivatives, then $y_n = nC_0 u_n v + nC_1 u_{n-1} v_1 + nC_2 u_{n-2} v_2 + nC_3 u_{n-3} v_3 + \dots + nC_n u v_n$

Or $y_n = nC_0 u v_n + nC_1 u_1 v_{n-1} + nC_2 u_2 v_{n-2} + nC_3 u_3 v_{n-3} + \dots + nC_n u_n v$

Proof : The result is proved by the principles of mathematical induction.

If $n = 1$ LHS = $y_1 = uv_1 + u_1 v$ RHS = $1C_0 u_1 v + 1C_1 u_0 v_1 = u_1 v + uv_1$

LHS = RHS. The result is true for $n = 1$.

Assume that the result is true for $n = m$

$$y_m = mC_0 u_m v + mC_1 u_{m-1} v_1 + mC_2 u_{m-2} v_2 + mC_3 u_{m-3} v_3 + \dots + mC_m u v_m$$

differentiating this equation w.r. to x

$$y_{m+1} = mC_0 (u_m v_1 + u_{m+1} v) + mC_1 (u_{m-1} v_2 + u_m v_1) + mC_2 (u_{m-2} v_3 + u_{m-1} v_2) + \dots + mC_m (u v_{m+1} + u_1 v_m)$$

$$y_{m+1} = mC_0 u_{m+1} v + (mC_1 + mC_0) u_m v_1 + (mC_2 + mC_1) u_{m-1} v_2 + \dots + mC_m u v_{m+1}$$

$$[\text{We have } mC_0 = (m+1)C_0, mC_m = (m+1)C_{m+1}, mC_r + mC_{r-1} = (m+1)C_r]$$

$$y_{m+1} = (m+1)C_0 u_{m+1} v + (m+1)C_1 u_{(m+1)-1} v_1 + \dots + (m+1)C_{m+1} u v_{m+1}$$

The result is true for $n = m + 1$

By mathematical induction the result is true for all positive integral values of n .

13. If $y = (\sin^{-1} x)^2$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$

Differentiating w.r.t. x $y_1 = \frac{2 \sin^{-1} x}{\sqrt{1 - x^2}} \Rightarrow \sqrt{1 - x^2} y_1 = 2 \sin^{-1} x$

again differentiating w.r.t. x $\sqrt{1 - x^2} y_2 + \frac{1}{2\sqrt{1 - x^2}} (-2x)y_1 = \frac{2}{\sqrt{1 - x^2}}$

$\Rightarrow (1 - x^2)y_2 - xy_1 = 2$,

Differentiating w.r.t. x n times by Leibnitz's theorem

$(1 - x^2)y_{n+2} + n(-2x)y_{n+1} + \frac{n(n-1)}{2}(-2)y_n - (xy_{n+1} + n(1)y_n) = 0$

$(1 - x^2)y_{n+2} - x(2n + 1)y_{n+1} + (-n^2 + n - n)y_n = 0$

$(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$

14. If $y = (\sin^{-1} x)$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$

Solution : Given, $y = \sin^{-1} x$ (1)

Diff. w.r.t. x , $y_1 = \frac{1}{\sqrt{1 - x^2}}$ (2) $\Rightarrow \sqrt{1 - x^2} y_1 = 1$

Diff. w.r.t. x , $\sqrt{1 - x^2} y_2 + y_1 \frac{1}{2\sqrt{1 - x^2}} (-2x) = 0$

Multiplying by $\sqrt{1 - x^2}$, $(1 - x^2)y_2 - xy_1 = 0$ (3)

Differentiating w.r.t. x , n times using Leibnitz theorem,

We get, $(1 - x^2)y_{n+2} + n(-2x)y_{n+1} + \frac{n(n-1)}{2}(-2)y_n - xy_{n+1} - n(1)y_n = 0$

i.e. $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (-n^2 + n - n)y_n = 0$

$(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$ (4)

15. If $y = (\tan^{-1} x)$, show that $(1 + x^2)y_{n+2} + 2(n + 1)xy_{n+1} + (n^2 + n)y_n = 0$

Given, $y = \tan^{-1} x$ (1)

Diff. w.r.t. x , $y_1 = \frac{1}{1 + x^2}$ (2) $\Rightarrow (1 + x^2)y_1 = 1$

Diff. w.r.t. x , $(1 + x^2)y_2 + 2xy_1 = 0$ (3)

Differentiating w.r.t. x , n times using Leibnitz theorem,

We get, $(1 + x^2)y_{n+2} + 2nxy_{n+1} + (n^2 - n)y_n + 2xy_{n+1} + 2ny_n = 0$

i.e. $(1 + x^2)y_{n+2} + 2(n + 1)xy_{n+1} + (n^2 + n)y_n = 0$

16. If $y = \cos(m \cos^{-1} x)$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - m^2)y_n = 0$

Solution : Given, $y = \cos(m \cos^{-1} x)$ (1)

Diff. w. r. t. x , $y_1 = -\sin(m \cos^{-1} x) \left(\frac{-m}{\sqrt{1-x^2}} \right)$

$$\Rightarrow \sqrt{(1-x^2)} y_1 = m \sin(m \cos^{-1} x)$$

Squaring, $(1-x^2) y_1^2 = m^2 \sin^2(m \cos^{-1} x)$

$$\text{i.e. } (1-x^2) y_1^2 = m^2 (1 - \cos^2(m \cos^{-1} x))$$

$$(1-x^2) y_1^2 = m^2 (1-y^2) \quad [\text{use (1)}]$$

Diff. w. r. t. x , $(1-x^2) 2 y_1 y_2 + y_1^2 (-2x) = m^2 (-2yy_1)$

Dividing by $2y_1$, $(1-x^2) y_2 - x y_1 = -m^2 y$

$$\Rightarrow (1-x^2) y_2 - x y_1 + m^2 y = 0 \quad \text{.....(2)}$$

Diff. w. r. t. x , n times using Leibnitz theorem,

$$(1-x^2) y_{n+2} + {}^n C_1 (-2x) y_{n+1} + {}^n C_2 (-2) y_n - x y_{n+1} - {}^n C_1 (1) y_n + m^2 y_n = 0$$

$$\text{i.e. } (1-x^2) y_{n+2} - 2nx y_{n+1} - \frac{2n(n-1)}{2} y_n - x y_{n+1} - n y_n + m^2 y_n = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - (2n+1)x y_{n+1} - (n^2 - n + n - m^2) y_n = 0$$

$$\therefore (1-x^2) y_{n+2} - (2n+1)x y_{n+1} - (n^2 - m^2) y_n = 0$$

17. If $x = \sin t$ and $y = \cos(pt)$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - p^2)y_n = 0$

Solution : Given, $x = \sin t$, $y = \cos pt$

$$\therefore \frac{dx}{dt} = \cos t, \quad \frac{dy}{dt} = -p \sin pt$$

$$y_1 = \frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{-p \sin pt}{\cos t}$$

$$y_1^2 = p^2 \frac{(\sin^2 pt)}{\cos^2 t} = p^2 \frac{(1 - \cos^2 pt)}{(1 - \sin^2 t)} = p^2 \frac{(1 - y^2)}{(1 - x^2)}$$

$$\text{or } (1-x^2) y_1^2 - p^2 (1-y^2) = 0$$

Differentiate this w. r. t. x ,

$$\text{we get } (1-x^2) 2 y_1 y_2 - 2x y_1^2 + 2p^2 y y_1 = 0$$

$$\text{or } (1-x^2) y_2 - x y_1 + p^2 y = 0$$

Differentiating this equation n times by using Leibnitz's rule

$$(1-x^2) y_{n+2} + n (-2x) y_{n+1} + \frac{n(n-1)}{2} (-2) y_n - (x y_{n+1} + n y_n) + p^2 y_n = 0$$

$$(1-x^2) y_{n+2} - 2x n y_{n+1} - n(n-1) y_n - x y_{n+1} - n y_n + p^2 y_n = 0$$

$$\therefore (1-x^2) y_{n+2} - (2n+1)x y_{n+1} - (n^2 - p^2) y_n = 0$$

18. If $y = \sin(m \sin^{-1} x)$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (m^2 - n^2)y_n = 0$

Given, $y = \sin(m \sin^{-1} x)$(1)

Diff. w.r.t. x , $y_1 = \cos(m \sin^{-1} x) \frac{m}{\sqrt{1-x^2}}$

$$\Rightarrow \sqrt{1-x^2}y_1 = m \cos(m \sin^{-1} x)$$

Squaring, $(1-x^2)y_1^2 = m^2 \sin^2(m \cos^{-1} x)$

i.e. $(1-x^2)y_1^2 = m^2(1 - \sin^2(m \cos^{-1} x))$

$$(1-x^2)y_1^2 = m^2(1-y^2) \quad [\text{use (1)}]$$

Diff. w.r.t. x , $(1-x^2)2y_1y_2 + y_1^2(-2x) = m^2(-2yy_1)$

Dividing by $2y_1$, $(1-x^2)y_2 - xy_1 = -m^2y$

$$\Rightarrow (1-x^2)y_2 - xy_1 + m^2y = 0 \dots\dots\dots(2)$$

Diff. w.r.t. x , n times using Leibnitz theorem,

$$(1-x^2)y_{n+2} - 2nxy_{n+1} + (n^2-n)y_n - xy_{n+1} - ny_n + m^2y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-n+n-m^2)y_n = 0$$

$$\therefore (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-m^2)y_n = 0$$

19. If $y = e^{\tan^{-1} x}$, show that $(1+x^2)y_{n+2} + [2(n+1)x-1]y_{n+1} + (n^2+n)y_n = 0$

Solution : Given $y = e^{\tan^{-1} x}$ (1)

Diff. w. r. t. x , $y_1 = e^{\tan^{-1} x} \frac{1}{(x^2+1)}$ (2) $\Rightarrow (x^2+1)y_1 = e^{\tan^{-1} x}$

Diff. w. r. t. x , $(x^2+1)y_2 + 2xy_1 = e^{\tan^{-1} x} \frac{1}{(1+x^2)}$

i.e., $(x^2+1)y_2 + 2xy_1 = y_1$ [use (2)]

$$\Rightarrow (x^2+1)y_2 + (2x-1)y_1 = 0 \dots\dots\dots(3)$$

Diff. n times w. r. t. x , using Leibnitz rule we get,

$$(x^2+1)y_{n+2} + n(2x)y_{n+1} + \frac{n(n-1)}{2}(2)y_n + (2x-1)y_{n+1} + n(2)y_n = 0$$

$$\text{i.e., } (x^2+1)y_{n+2} + [2(n+1)x-1]y_{n+1} + (n^2-n+2n)y_n = 0$$

$$\Rightarrow (x^2+1)y_{n+2} + [2(n+1)x-1]y_{n+1} + (n^2+n)y_n = 0$$

$$\Rightarrow (x^2+1)y_{n+2} + [2(n+1)x-1]y_{n+1} + n(n+1)y_n = 0$$

In this result replace n by $n-2$, we get

$$(x^2+1)y_n + \{2[(n-2)+1]x-1\}y_{n-1} + (n-2)(n-2+1)y_{n-2} = 0$$

$$\text{i.e. } (x^2+1)y_n + \{2(n-1)x-1\}y_{n-1} + (n-2)(n-1)y_{n-2} = 0$$

20. If $y = e^{m \sin^{-1} x}$, show that $(1 - x^2)y_{n+2} + (2n + 1)xy_{n+1} - (n^2 - a^2)y_n = 0$

Solution : Given $y = e^{m \sin^{-1} x}$, Diff. w. r. t. x , $y_1 = e^{m \sin^{-1} x} \left(\frac{m}{\sqrt{1-x^2}} \right) = \frac{my}{\sqrt{1-x^2}}$

$$\sqrt{1-x^2} y_1 = my$$

Squaring both sides $(1-x^2) y_1^2 = m^2 y^2$

Differentiating again w. r. t. x , $(1-x^2) 2y_1 y_2 + y_1^2 (-2x) = m^2 2y y_1$

$$\Rightarrow (1-x^2) y_2 - x y_1 = m^2 y \text{ or } (1-x^2) y_2 - x y_1 - m^2 y = 0$$

Differentiating this equation n times by using Leibnitz rule, we get

$$(1-x^2) y_{n+2} + n(-2x) y_{n+1} + \frac{n(n-1)}{2} (-2) y_n - (x y_{n+1} + n y_n) - m^2 y_n = 0$$

$$(1-x^2) y_{n+2} - 2nx y_{n+1} - n(n-1) y_n - x y_{n+1} - n y_n - m^2 y_n = 0$$

21. If $y = a \cos(\log x) + b \sin(\log x)$, show that $x^2 y_{n+2} + (2n + 1)xy_{n+1} + (n^2 + 1)y_n = 0$

Solution : We have $y = a \cos(\log x) + b \sin(\log x)$

$$\therefore y_1 = -\frac{a}{x} \sin(\log x) + \frac{b}{x} \cos(\log x) \text{ or } xy_1 = -a \sin(\log x) + b \cos(\log x)$$

Differentiating w.r.t x , again we get $xy_2 + y_1 = -\frac{a}{x} \cos(\log x) - \frac{b}{x} \sin(\log x)$

$$x^2 y_2 + xy_1 = -y \text{ or } x^2 y_2 + xy_1 + y = 0$$

Applying Leibnitz's theorem, we have

$${}^n C_0 x^2 y_{n+2} + {}^n C_1 y_{n+1} 2x + {}^n C_2 y_n \cdot 2 + {}^n C_0 xy_{n+1} + {}^n C_1 y_n + y_n = 0$$

$$x^2 y_{n+2} + ny_{n+1} 2x + \frac{n(n-1)}{2} y_n \cdot 2 + xy_{n+1} + ny_n + y_n = 0$$

$$x^2 y_{n+2} + xy_{n+1} (2n+1) + (n^2 + 1)y_n = 0$$

22. If $z = \frac{xy}{y-x}$ then verify that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$

Solution : (i) $z = \frac{xy}{y-x}$ (1)

Diff. (1) partially w. r. t. x , $\frac{\partial z}{\partial x} = \frac{(y-x)(y) - xy(-1)}{(y-x)^2} = \frac{y^2 - xy + xy}{(y-x)^2}$

$\therefore \frac{\partial z}{\partial x} = \frac{y^2}{(y-x)^2}$ (2)

Diff. (2) partially w. r. t. y , $\frac{\partial^2 z}{\partial y \partial x} = \frac{(y-x)^2 (2y) - y^2 (2)(y-x)(1)}{(y-x)^4}$
 $= \frac{2y(y-x)[y-x-y]}{(y-x)^4}$

$\frac{\partial^2 z}{\partial y \partial x} = \frac{-2xy}{(y-x)^3}$ (3)

Diff. (1) partially w. r. t. y , $\frac{\partial z}{\partial y} = \frac{(y-x)x - xy(1)}{(y-x)^2} = \frac{xy - x^2 - xy}{(y-x)^2}$

$\frac{\partial z}{\partial y} = -\frac{x^2}{(y-x)^2}$ (4)

Diff (4) partially w. r. t. x , $\frac{\partial^2 z}{\partial x \partial y} = -\frac{(y-x)^2 (2x) - x^2 2(y-x)(-1)}{(y-x)^4}$
 $= -\frac{2x(y-x)[y-x+x]}{(y-x)^4}$

$\frac{\partial^2 z}{\partial x \partial y} = -\frac{2xy}{(y-x)^3}$ (5)

From (4) and (5) we find that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$

23. If $z = \sin(ax + y) + \cos(ax - y)$ then verify that $\frac{\partial^2 z}{\partial x^2} = a^2 \frac{\partial^2 z}{\partial y^2}$

Solution : Given $Z = \sin(ax+y) + \cos(ax-y)$ (1)

Diff (1) w. r. t. x , partially, $\frac{\partial z}{\partial x} = \cos(ax+y) a - \sin(ax-y) a$ (2)

Diff (2) w. r. t. x , partially, $\frac{\partial^2 z}{\partial x^2} = -\sin(ax+y) a^2 - \cos(ax-y) a^2$
 $= -a^2 [\sin(ax+y) + \cos(ax-y)]$

$\therefore \frac{\partial^2 z}{\partial x^2} = -a^2 z$ (3)

Diff (1) w. r. t. y , partially, $\frac{\partial z}{\partial y} = \cos(ax+y) (1) - \sin(ax-y) (-1)$
 $= \cos(ax+y) + \sin(ax-y)$ (4)

Diff (4) w. r. t. y , partially, $\frac{\partial^2 z}{\partial y^2} = -\sin(ax+y) (1) + \cos(ax-y) (-1)$
 $= -[\sin(ax+y) + \cos(ax-y)]$

$\therefore \frac{\partial^2 z}{\partial y^2} = -z$ (5)

From (3) and (5) we observe that $\frac{\partial^2 z}{\partial x^2} = a^2 \frac{\partial^2 z}{\partial y^2}$

24. If $u = \varphi(y + ax) + \psi(y - ax)$ then verify that $\frac{\partial^2 u}{\partial x^2} = a^2 \frac{\partial^2 u}{\partial y^2}$

We have $u = \varphi(y + ax) + \psi(y - ax)$(1)

Differentiating (1) partially w.r.t x , we get

$$\begin{aligned}\frac{\partial u}{\partial x} &= \varphi'(y + ax) \frac{\partial}{\partial x}(y + ax) + \psi'(y - ax) \frac{\partial}{\partial x}(y - ax) \\ &= \varphi'(y + ax) \cdot a - a\psi'(y - ax)\end{aligned}$$

And again, differentiating partially w.r.t x , we get

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= a^2 \varphi''(y + ax) + a^2 \psi''(y - ax) \\ &= a^2 (\varphi''(y + ax) + \psi''(y - ax))\end{aligned}\text{.....(2)}$$

Differentiating (1) partially w.r.t y , we get

$$\begin{aligned}\frac{\partial u}{\partial y} &= \varphi'(y + ax) \frac{\partial}{\partial y}(y + ax) + \psi'(y - ax) \frac{\partial}{\partial y}(y - ax) \\ &= \varphi'(y + ax) \cdot 1 + 1 \cdot \psi'(y - ax)\end{aligned}$$

And again, differentiating partially w.r.t y , we get

$$\frac{\partial^2 u}{\partial y^2} = \varphi''(y + ax) + \psi''(y - ax)\text{.....(3)}$$

From (2) and (3) it follows that

$$\frac{\partial^2 u}{\partial x^2} = a^2 \frac{\partial^2 u}{\partial y^2}$$

25. If $z = e^x(x \sin y + y \cos y)$, Prove that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$

Solution : Given $z = e^x(x \sin y + y \cos y)$ (1)

$$\begin{aligned}\text{Diff. (1) w. r. t. } x, \text{ partially, } \frac{\partial z}{\partial x} &= e^x (\sin y) + (x \sin y + y \cos y) e^x \\ &= e^x (\sin y + x \sin y + y \cos y) \end{aligned}\text{.....(2)}$$

$$\begin{aligned}\text{Diff. (2) w. r. t. } x, \text{ partially, } \frac{\partial^2 z}{\partial x^2} &= e^x (\sin y) + (\sin y + x \sin y + y \cos y) e^x \\ &= e^x (2 \sin y + x \sin y + y \cos y) \end{aligned}\text{.....(3)}$$

$$\text{Diff. (1) w. r. t. } y, \text{ partially, } \frac{\partial z}{\partial y} = e^x (x \cos y - y \sin y + \cos y) \text{.....(4)}$$

$$\begin{aligned}\text{Diff. (4) w. r. t. } y, \text{ partially, } \frac{\partial^2 z}{\partial y^2} &= e^x (-x \sin y - y \cos y - \sin y - \sin y) \\ &= -e^x (2 \sin y + x \sin y + y \cos y) \end{aligned}\text{.....(5)}$$

$$\text{Adding (4) and (5) we get } \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

26. If $z = (1 - 2xy + y^2)^{-1/2}$, show that $x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = y^2 z^3$

Given $z = (1 - 2xy + y^2)^{-1/2}$(1)

Diff (1) w.r.t x , partially, $\frac{\partial z}{\partial x} = \frac{-1}{2} (1 - 2xy + y^2)^{-3/2} \cdot -2y$

$$= y(1 - 2xy + y^2)^{-3/2} = yz^3$$

Diff (1) w.r.t y , partially, $\frac{\partial z}{\partial y} = \frac{-1}{2} (1 - 2xy + y^2)^{-3/2} \cdot (-2x + 2y)$

$$= (x - y)(1 - 2xy + y^2)^{-3/2} = (x - y)z^3$$

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = xyz^3 - (x - y)z^3 = yz^3$$

27. If $u = x^2y + y^2z + z^2x$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = (x + y + z)^2$

Given $u = x^2y + y^2z + z^2x$(1)

Diff (1) w.r.t x , partially, $\frac{\partial u}{\partial x} = 2xy + z^2$

Diff (1) w.r.t y , partially, $\frac{\partial u}{\partial y} = 2yz + x^2$

Diff (1) w.r.t z , partially, $\frac{\partial u}{\partial z} = 2xz + y^2$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 2xy + z^2 + 2yz + x^2 + 2xz + y^2 = (x + y + z)^2$$

28. If $u = (x - y)^n + (y - z)^n + (z - x)^n$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Given $u = (x - y)^n + (y - z)^n + (z - x)^n$(1)

Diff (1) w.r.t x , partially, $\frac{\partial u}{\partial x} = n(x - y)^{n-1} - n(z - x)^{n-1}$

Diff (1) w.r.t y , partially, $\frac{\partial u}{\partial y} = n(y - z)^{n-1} - n(x - y)^{n-1}$

Diff (1) w.r.t z , partially, $\frac{\partial u}{\partial z} = n(z - x)^{n-1} - n(y - z)^{n-1}$

$$\begin{aligned} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} &= n(x - y)^{n-1} - n(z - x)^{n-1} + n(y - z)^{n-1} - n(x - y)^{n-1} \\ &\quad + n(z - x)^{n-1} - n(y - z)^{n-1} = 0 \end{aligned}$$

29. If $u = r^n$ where $r^2 = x^2 + y^2 + z^2$ show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = n(n+1)r^{n-2}$

Solution : Given $u = r^n$ where $r^2 = x^2 + y^2 + z^2$. Here u is a function of r and r is a function of x, y and z . Hence u is a composite function of x, y, z . We shall use the chain rule of differentiation.

$$r^2 = x^2 + y^2 + z^2$$

Diff w. r. t. x , $2r \frac{\partial r}{\partial x} = 2x \quad \therefore \frac{\partial r}{\partial x} = \frac{x}{r}$

Similarly $\frac{\partial r}{\partial y} = \frac{y}{r}$ and $\frac{\partial r}{\partial z} = \frac{z}{r}$

$$u = r^n$$

Diff. partially w. r. t. x , $\frac{\partial u}{\partial x} = \frac{\partial}{\partial r} (r^n) \frac{\partial r}{\partial x} = nr^{n-1} \frac{\partial r}{\partial x}$

$$\therefore \frac{\partial u}{\partial x} = nr^{n-1} \frac{x}{r} = nr^{n-2} x \dots\dots\dots(1)$$

Diff. (1) w. r. t. x , partially, $\frac{\partial^2 u}{\partial x^2} = n \frac{\partial}{\partial x} (r^{n-2} x)$

$$= nx \frac{\partial}{\partial x} (r^{n-2}) + nr^{n-2} \frac{\partial}{\partial x} (x) \quad [\text{by product rule}]$$

$$= nx \frac{\partial}{\partial r} (r^{n-2}) \frac{\partial r}{\partial x} + r^{n-2} (1) \quad [\text{by chain rule}]$$

$$= nx(n-2)r^{n-3} \frac{x}{r} + nr^{n-2}$$

$$\frac{\partial^2 u}{\partial x^2} = n(n-2)r^{n-4} x^2 + nr^{n-2} \dots\dots\dots(2)$$

Similarly $\frac{\partial^2 u}{\partial y^2} = n(n-2)r^{n-4} y^2 + nr^{n-2} \dots\dots\dots(3)$

and $\frac{\partial^2 u}{\partial z^2} = n(n-2)r^{n-4} z^2 + nr^{n-2} \dots\dots\dots(4)$

Adding (2), (3) and (4) we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = n(n-2)r^{n-4} (x^2 + y^2 + z^2) + 3r^{n-2}$$

$$= n(n-2)r^{n-4} r^2 + 3nr^{n-2}$$

$$= n(n-2)r^{n-2} + 3nr^{n-2} = n[n-2+3]r^{n-2} = n(n+1)r^{n-2}$$

30. If $u = f(r)$ where $r = \sqrt{x^2 + y^2 + z^2}$ show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r} f'(r)$

Solution : We have $r^2 = x^2 + y^2 + z^2$

$$2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r} \dots\dots\dots(1)$$

Differentiating $u = f(r)$ partially w. r. t. x , we get

$$\frac{\partial u}{\partial x} = f'(r) \frac{\partial r}{\partial x} = f'(r) \frac{x}{r} \quad \text{using (1)}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(f'(r) \frac{x}{r} \right) = \frac{d}{dr} \left(\frac{f'(r)}{r} \right) \frac{\partial r}{\partial x} x + \frac{f'(r)}{r} = \left[\frac{f''(r)}{r} - \frac{f'(r)}{r^2} \right] \frac{x}{r} x + \frac{f'(r)}{r}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{x^2}{r^2} f''(r) + \frac{f'(r)}{r} \left[1 - \frac{x^2}{r^2} \right] \dots\dots\dots(2)$$

Similarly $\frac{\partial^2 u}{\partial y^2} = \frac{y^2}{r^2} f''(r) + \frac{f'(r)}{r} \left[1 - \frac{y^2}{r^2} \right] \dots\dots\dots(3)$

$$\frac{\partial^2 u}{\partial z^2} = \frac{z^2}{r^2} f''(r) + \frac{f'(r)}{r} \left[1 - \frac{z^2}{r^2} \right] \dots\dots\dots(4)$$

Adding (2), (3) and (4) we get,

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} &= \frac{x^2 + y^2 + z^2}{r^2} f''(r) + \frac{f'(r)}{r} \left[3 - \frac{x^2 + y^2 + z^2}{r^2} \right] \\ &= f''(r) + \frac{2}{r} f'(r) \quad [x^2 + y^2 + z^2 = r^2] \end{aligned}$$

31. If $u = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$ show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

Solution: (a) We have $u = (x^2 + y^2 + z^2)^{-1/2} \dots\dots\dots(1)$

Differentiating (1) partially w.r.t. x , we get,

$$\begin{aligned} \frac{\partial u}{\partial x} &= -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2x \\ &= -x(x^2 + y^2 + z^2)^{-3/2} \dots\dots\dots(2) \end{aligned}$$

Similarly, $\frac{\partial u}{\partial y} = -y(x^2 + y^2 + z^2)^{-3/2}; \quad \frac{\partial u}{\partial z} = -z(x^2 + y^2 + z^2)^{-3/2}$

Hence, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = -(x^2 + y^2 + z^2)^{-3/2} (x + y + z) = -(x^2 + y^2 + z^2)^{-1/2}$
 $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = -u \quad \text{(using (1))}$

(b) Differentiating (2) partially w.r.t. x , we get,

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(-x(x^2 + y^2 + z^2)^{-3/2} \right) \\ &= - \left[x \left(-\frac{3}{2} \right) (x^2 + y^2 + z^2)^{-5/2} 2x + 1 \cdot (x^2 + y^2 + z^2)^{-3/2} \right] \\ &= -(x^2 + y^2 + z^2)^{-5/2} [x^2 + y^2 + z^2 - 3x^2] \\ &= (x^2 + y^2 + z^2)^{-5/2} [2x^2 - y^2 - z^2] \dots\dots\dots(3) \end{aligned}$$

Similarly, $\frac{\partial^2 u}{\partial y^2} = (x^2 + y^2 + z^2)^{-5/2} [-x^2 + 2y^2 - z^2] \dots\dots\dots(4)$

$$\frac{\partial^2 u}{\partial z^2} = (x^2 + y^2 + z^2)^{-5/2} [-x^2 - y^2 + 2z^2] \dots\dots\dots(5)$$

Adding (3), (4) and (5) we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = (x^2 + y^2 + z^2)^{-5/2} \cdot 0 = 0$$

32. State and prove Euler's theorem for homogeneous functions

Theorem: If $u = f(x, y)$ is a homogeneous function of x and y with degree n , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

Proof: Since $u = f(x, y)$ is a homogeneous function of x and y with degree n .

Thus $u = f(x, y)$ can be written as $u = x^n \phi\left(\frac{y}{x}\right)$.

$$\begin{aligned} \text{This gives } \frac{\partial u}{\partial x} &= nx^{n-1} \phi\left(\frac{y}{x}\right) + x^n \phi'\left(\frac{y}{x}\right) \left(-\frac{y}{x^2}\right) \\ &= nx^{n-1} \phi\left(\frac{y}{x}\right) - x^{n-2} y \phi'\left(\frac{y}{x}\right) \end{aligned}$$

$$\text{and } \frac{\partial u}{\partial y} = x^n \phi'\left(\frac{y}{x}\right) \cdot \frac{1}{x} = x^{n-1} \phi'\left(\frac{y}{x}\right)$$

Therefore,

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= nx^n \phi\left(\frac{y}{x}\right) - x^{n-1} y \phi'\left(\frac{y}{x}\right) + x^{n-1} y \phi'\left(\frac{y}{x}\right) \\ &= nx^n \phi\left(\frac{y}{x}\right) = nu \end{aligned}$$

This completes the proof.

33. Verify Euler's theorem for the function $u = ax^2 + 2hxy + by^2$

Proof : i) $u = ax^2 + 2hxy + by^2$

$$u = x^2 \left[a + 2h\left(\frac{y}{x}\right) + b\left(\frac{y}{x}\right)^2 \right] = x^2 f\left(\frac{y}{x}\right). \text{ Hence } u \text{ is a homogeneous function of degree 2}$$

in x and y . Therefore according to Euler's theorem $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$. This is to be

$$\text{verified. } \frac{\partial u}{\partial x} = 2ax + 2hy, \quad \frac{\partial u}{\partial y} = 2hx + 2by$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2ax^2 + 2hxy + 2hxy + 2by^2 = 2(ax^2 + 2hxy + by^2) = 2u$$

Thus Euler's theorem is verified.

34. Verify Euler's theorem for the function $u = x^n \log\left(\frac{y}{x}\right)$

$$(iii) u = x^n \log\left(\frac{y}{x}\right)$$

$$u(\lambda x, \lambda y) = \lambda^n x^n \log\left(\frac{\lambda y}{\lambda x}\right)$$

$$\Rightarrow u(\lambda x, \lambda y) = \lambda^n x^n \log\left(\frac{y}{x}\right) = \lambda^n u(x, y)$$

Hence u is a homogeneous function of degree n in x and y . Therefore according to Euler's theorem $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$. This is to be verified.

$$\frac{\partial u}{\partial x} = x^n \cdot \frac{x}{y} \left(-\frac{y}{x^2}\right) + \log\left(\frac{y}{x}\right) n x^{n-1}$$

$$\Rightarrow x \frac{\partial u}{\partial x} = -x^n + \log\left(\frac{y}{x}\right) n x^n \quad \dots\dots\dots (1)$$

$$\frac{\partial u}{\partial y} = x^n \cdot \frac{x}{y} \left(\frac{1}{x}\right) = \frac{x^n}{y}$$

$$\Rightarrow y \frac{\partial u}{\partial y} = x^n \quad \dots\dots\dots (2)$$

$$\text{Adding (1) and (2) we get, } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -x^n + n x^n \log\left(\frac{y}{x}\right) + x^n = n x^n \log\left(\frac{y}{x}\right)$$

i.e. $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$. Thus Euler's theorem is verified

35. Verify Euler's theorem for the function $u = x^4 \log \left(\frac{y}{x} \right)$

$$(iv) u = x^4 \log \left(\frac{y}{x} \right) = x^4 \phi \left(\frac{y}{x} \right)$$

Hence u is a homogeneous function of degree 4 in x and y .

To verify Euler's theorem, we have to prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 4u$

$$\text{Now, } u = x^4 \log \left(\frac{y}{x} \right) = x^4 (\log y - \log x)$$

Differentiating (1) partially w.r.t. x , we get

$$\frac{\partial u}{\partial x} = 4x^3 (\log y - \log x) + x^4 \left(-\frac{1}{x} \right)$$

$$\therefore x \frac{\partial u}{\partial x} = 4x^4 (\log y - \log x) - x^4$$

Also, differentiating (1) partially w.r.t. y , we get

$$\frac{\partial u}{\partial y} = x^4 \frac{1}{y}$$

$$\therefore y \frac{\partial u}{\partial y} = x^4$$

Adding (2) and (3), we get

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 4x^4 (\log y - \log x) = 4u$$

Hence the verification.

36. If

$$u = \sin^{-1} \left(\frac{x^2 - y^2}{x + y} \right),$$

show

that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$$

Example

Solution : Here u is not a homogeneous function, we however write

$$z = \sin u = \frac{x^2 + y^2}{x + y} = \frac{x^2 \left(1 + \left(\frac{y}{x} \right)^2 \right)}{x \left(1 + \left(\frac{y}{x} \right) \right)}$$

So that z is a homogeneous function of x, y of degree 1.

$$\therefore x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 1 \cdot z \dots\dots\dots (1)$$

$$\text{But } \frac{\partial z}{\partial x} = \cos u \frac{\partial u}{\partial x}; \quad \frac{\partial z}{\partial y} = \cos u \frac{\partial u}{\partial y}$$

$$\text{Substituting in (1), we obtain, } \cos u \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = 1 \cdot z = \sin u$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$$

37. If $u = \cos^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -2 \cot u$

$$\text{Proof: } u = \cos^{-1} \left(\frac{x^3 + y^3}{x + y} \right) \Rightarrow \cos u = \frac{x^3 + y^3}{x + y} \dots\dots\dots (1)$$

$$\text{Let, } z = \cos u = \frac{x^3 + y^3}{x + y} \dots\dots\dots (2)$$

z is a homogeneous function of degree 2 in x and y .

$$\text{Hence by Euler's theorem, } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 2z \dots\dots\dots (3)$$

$$\text{Since } z = \cos u, \quad \frac{\partial z}{\partial x} = -\sin u \frac{\partial u}{\partial x} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\sin u \frac{\partial u}{\partial y}$$

Substituting these in (3)

$$x \left(-\sin u \frac{\partial u}{\partial x} \right) + y \left(-\sin u \frac{\partial u}{\partial y} \right) = 2 \cos u$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -2 \cot u. \quad \text{This proves the first part of 1}$$

38. If $u = \tan^{-1} \left(\frac{x^3+y^3}{x-y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$

Solution : $u = \tan^{-1} \left(\frac{x^3+y^3}{x-y} \right)$

$$\Rightarrow \tan u = \frac{x^3+y^3}{x-y} = \frac{x^3 \left(1 + \left(\frac{y}{x} \right)^3 \right)}{x \left(1 - \left(\frac{y}{x} \right) \right)} = \frac{x^2 \left(1 + \left(\frac{y}{x} \right)^3 \right)}{\left(1 - \left(\frac{y}{x} \right) \right)} = x^2 \phi \left(\frac{y}{x} \right) = z,$$

So that z is a homogeneous function of x, y of degree 2.

By Euler's theorem $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{nf(u)}{f'(u)} = \frac{2 \tan u}{\sec^2 u} = \frac{2 \sin u \cos^2 u}{\cos u}$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

39. If $u = \log \left(\frac{x^4+y^4}{x-y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$

Solution: $u = \log \left(\frac{x^4+y^4}{x-y} \right) = \log_e \left(\frac{x^4+y^4}{x-y} \right)$

Therefore, $e^u = \frac{x^4+y^4}{x-y} = \frac{x^4(1+(y/x)^4)}{x(1-(y/x))} = x^3 \phi \left(\frac{y}{x} \right)$

e^u is a homogeneous function of degree 3 in x and y .

Applying Euler's theorem for the function e^u we have,

$$x \frac{\partial(e^u)}{\partial x} + y \frac{\partial(e^u)}{\partial y} = ne^u \quad ; \text{ here } n = 3$$

$$\Rightarrow x e^u \frac{\partial u}{\partial x} + y e^u \frac{\partial u}{\partial y} = 3 e^u$$

Dividing both sides by e^u , we have

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$$

40. If $u = \log \left(\frac{x^3 + x^2y - y^2x + 2y^3}{x+y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2$

Solution: Given, $u = \log \left(\frac{x^3 + x^2y - y^2x + 2y^3}{x+y} \right)$

$\Rightarrow \frac{x^3 + x^2y - y^2x + 2y^3}{x+y} = e^u = z$, a homogeneous function of degree 2 in x and y .

$\therefore$ By Euler's theorem, $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 2z$ (1)

Since $z = e^u$, $\frac{\partial z}{\partial x} = e^u \frac{\partial u}{\partial x}$ and $\frac{\partial z}{\partial y} = e^u \frac{\partial u}{\partial y}$

Substituting in (1) $x \left(e^u \frac{\partial u}{\partial x} \right) + y \left(e^u \frac{\partial u}{\partial y} \right) = 2e^u$

$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2$ (2)

41. If Find $\frac{dz}{dt}$ if $z = xy^2 + x^2y$ where $x = at^2$ and $y = 2at$

Solution: Here, $z = xy^2 + x^2y$ is function of x and y and x and y are functions of t .

Therefore, the total derivative $\frac{dz}{dt}$ is given by the formula

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \quad \dots \dots (1)$$

From given $z = xy^2 + x^2y$, we find that

$$\frac{\partial z}{\partial x} = y^2 + 2xy; \quad \frac{\partial z}{\partial y} = 2xy + x^2$$

From given $x = at^2$ and $y = 2at$, we find that

$$\frac{dx}{dt} = 2at; \quad \frac{dy}{dt} = 2a$$

Substituting these into (1), we get

$$\begin{aligned} \frac{dz}{dt} &= (y^2 + 2xy)2at + (2xy + x^2)2a \\ &= y(y^2 + 2xy) + 2a(2xy + x^2) \quad \text{using } y = 2at \\ &= y^3 + 2xy^2 + 4axy + 2ax^2 \end{aligned}$$

This gives $\frac{dz}{dt}$ in terms of x and y . By substituting for x and y , we can rewrite this in terms of t .

Thus

$$\begin{aligned} \frac{dz}{dt} &= (2at)^3 + 2(at^2)(2at)^2 + 4a(at^2)(2at) + 2a(at^2)^2 \\ \frac{dz}{dt} &= 2a^3(8t^3 + 5t^4) \quad \dots \dots (2) \end{aligned}$$

42. Find the total derivative of u w.r.t ' t ' when $u = e^x \sin y$ where $x = \log t$ and $y = t^2$

Here u is a function of x, y and x, y are functions of t .

i.e., $\{u \rightarrow (x, y, z) \rightarrow t\} \Rightarrow u \rightarrow t \quad \therefore \frac{du}{dt}$ is the total derivative.

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} = e^x \sin y \left(\frac{1}{t}\right) + e^x \cos y (2t) = t \sin t^2 \left(\frac{1}{t}\right) + t \cos(t^2)(2t) \\ &= \sin t^2 + 2t^2 \cos t^2 \end{aligned}$$

43. Find $\frac{dz}{dt}$ if $u = x^2 + y^2$ where $x = \frac{1}{t}$ and $y = t^2$.

Here u is a function of x, y and x, y are functions of t .

i.e., $\{u \rightarrow (x, y, z) \rightarrow t\} \Rightarrow u \rightarrow t \quad \therefore \frac{du}{dt}$ is the total derivative.

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} = 2x \left(\frac{-1}{t^2}\right) + 2y(2t) = 2 \left(\frac{1}{t}\right) \left(\frac{-1}{t^2}\right) + 2t^2(2t) = \frac{-2}{t^3} + 4t^3$$

44. If $z = e^{ax+by} f(ax - by)$ show that $b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2abz$

Solution: We have $z = e^{ax+by} f(ax - by)$

Let $u = ax + by; v = ax - by$, we find

$$\frac{\partial u}{\partial x} = a; \frac{\partial u}{\partial y} = b \quad \text{and} \quad \frac{\partial v}{\partial x} = a; \frac{\partial v}{\partial y} = -b$$

$$\text{so that } z = e^u f(v) \Rightarrow \frac{\partial z}{\partial u} = e^u f(v); \frac{\partial z}{\partial v} = e^u f'(v)$$

Hence $z \rightarrow (u, v) \rightarrow (x, y) \Rightarrow z \rightarrow (x, y)$

Thus z is a composite function of x and y , by using chain rules, we have

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} \\ &= e^u f(v) \cdot a + e^u f'(v) \cdot a \\ \frac{\partial z}{\partial x} &= ae^u [f(v) + f'(v)] \quad \dots \dots \dots (1)\end{aligned}$$

Again, we have

$$\begin{aligned}\frac{\partial z}{\partial y} &= \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} \\ &= e^u f(v) \cdot b + e^u f'(v) \cdot (-b) \\ \frac{\partial z}{\partial y} &= be^u [f(v) - f'(v)] \quad \dots \dots \dots (2)\end{aligned}$$

Multiply (1) by b and (2) by a and add, we get

$$\begin{aligned}b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} &= bae^u [f(v) + f'(v)] + a be^u [f(v) - f'(v)] \\ &= 2abe^u f(v) \\ b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} &= 2abz\end{aligned}$$

45. If $u = f(y - z, z - x, x - y)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Solution: Here, $u = f(s, t, w)$,

where, $s = y - z$; $t = z - x$; $w = x - y$

$$\frac{\partial s}{\partial x} = 0; \frac{\partial s}{\partial y} = 1; \frac{\partial s}{\partial z} = -1; \quad \frac{\partial t}{\partial x} = -1; \frac{\partial t}{\partial y} = 0; \frac{\partial t}{\partial z} = 1; \quad \frac{\partial w}{\partial x} = 1; \frac{\partial w}{\partial y} = -1; \frac{\partial w}{\partial z} = 0$$

Therefore,

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial x} + \frac{\partial u}{\partial w} \frac{\partial w}{\partial x} \\ &= \frac{\partial u}{\partial s} (0) + \frac{\partial u}{\partial t} (-1) + \frac{\partial u}{\partial w} \cdot 1 = -\frac{\partial u}{\partial t} + \frac{\partial u}{\partial w} \quad \dots \dots \dots (1)\end{aligned}$$

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y} + \frac{\partial u}{\partial w} \frac{\partial w}{\partial y} \\ &= \frac{\partial u}{\partial s} \cdot 1 + \frac{\partial u}{\partial t} (0) + \frac{\partial u}{\partial w} (-1) = \frac{\partial u}{\partial s} - \frac{\partial u}{\partial w} \quad \dots \dots \dots (2)\end{aligned}$$

$$\begin{aligned}\frac{\partial u}{\partial z} &= \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial z} + \frac{\partial u}{\partial w} \frac{\partial w}{\partial z} \\ &= \frac{\partial u}{\partial s} (-1) + \frac{\partial u}{\partial t} \cdot 1 + \frac{\partial u}{\partial w} (0) = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \quad \dots \dots \dots (3)\end{aligned}$$

Adding (1), (2) and (3), we get

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

46. If $u = u(r, s, t)$, where $r = \frac{x}{y}$, $s = \frac{y}{z}$, $t = \frac{z}{x}$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$

Solution: Given $r = \frac{x}{y}$, $s = \frac{y}{z}$, and $t = \frac{z}{x}$

$$\therefore \frac{\partial r}{\partial x} = \frac{1}{y}; \frac{\partial r}{\partial y} = -\frac{x}{y^2}; \frac{\partial r}{\partial z} = 0; \quad \frac{\partial s}{\partial x} = 0; \frac{\partial s}{\partial y} = \frac{1}{z}; \frac{\partial s}{\partial z} = -\frac{y}{z^2}; \quad \frac{\partial t}{\partial x} = -\frac{z}{x^2}; \frac{\partial t}{\partial y} = 0; \frac{\partial t}{\partial z} = \frac{1}{x};$$

Therefore,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial x} \\ &= \frac{\partial u}{\partial r} \left(\frac{1}{y} \right) + \frac{\partial u}{\partial s} (0) + \frac{\partial u}{\partial t} \left(-\frac{z}{x^2} \right) = \frac{1}{y} \frac{\partial u}{\partial r} - \frac{z}{x^2} \frac{\partial u}{\partial t} \\ \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y} \\ &= \frac{\partial u}{\partial r} \left(-\frac{x}{y^2} \right) + \frac{\partial u}{\partial s} \left(\frac{1}{z} \right) + \frac{\partial u}{\partial t} (0) = -\frac{x}{y^2} \frac{\partial u}{\partial r} + \frac{1}{z} \frac{\partial u}{\partial s} \\ \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial z} \\ &= \frac{\partial u}{\partial r} (0) + \frac{\partial u}{\partial s} \left(-\frac{y}{z^2} \right) + \frac{\partial u}{\partial t} \left(\frac{1}{x} \right) = -\frac{y}{z^2} \frac{\partial u}{\partial s} + \frac{1}{x} \frac{\partial u}{\partial t} \end{aligned}$$

From these expressions, we get

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} &= x \left(\frac{1}{y} \frac{\partial u}{\partial r} - \frac{z}{x^2} \frac{\partial u}{\partial t} \right) + y \left(-\frac{x}{y^2} \frac{\partial u}{\partial r} + \frac{1}{z} \frac{\partial u}{\partial s} \right) + z \left(-\frac{y}{z^2} \frac{\partial u}{\partial s} + \frac{1}{x} \frac{\partial u}{\partial t} \right) \\ &= \left(\frac{x}{y} - \frac{x}{y} \right) \frac{\partial u}{\partial r} + \left(\frac{y}{z} - \frac{y}{z} \right) \frac{\partial u}{\partial s} + \left(\frac{z}{x} - \frac{z}{x} \right) \frac{\partial u}{\partial t} \\ &= 0 \end{aligned}$$

47. If $u = f(x, y)$, where $x = r \cos \theta$ and $y = r \sin \theta$ show that

$$\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2$$

Solution: Here, $u \rightarrow (x, y) \rightarrow (r, \theta) \Rightarrow u \rightarrow (r, \theta)$

Therefore, by chain rules, we have

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial r} = \frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta \quad \dots \dots (1)$$

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial u}{\partial x} (-r \sin \theta) + \frac{\partial u}{\partial y} (r \cos \theta) \quad \dots \dots (2)$$

Squaring both sides of (1) and (2) we obtain

$$\left(\frac{\partial u}{\partial r} \right)^2 = \left(\frac{\partial u}{\partial x} \right)^2 \cos^2 \theta + \left(\frac{\partial u}{\partial y} \right)^2 \sin^2 \theta + 2 \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} \cos \theta \sin \theta$$

$$\left(\frac{\partial u}{\partial \theta} \right)^2 = \left(\frac{\partial u}{\partial x} \right)^2 r^2 \sin^2 \theta + \left(\frac{\partial u}{\partial y} \right)^2 r^2 \cos^2 \theta - 2 \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} r^2 \cos \theta \sin \theta$$

These give

$$\begin{aligned} \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2 &= \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] (\cos^2 \theta + \sin^2 \theta) \\ &= \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \end{aligned}$$

as required.

48. If $u = x + y + z, v = y + z, w = z$ show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 1$

We have $u = x + y + z, v = y + z, w = z$

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial u}{\partial y} = 1, \quad \frac{\partial u}{\partial z} = 1$$

$$\frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = 1, \quad \frac{\partial v}{\partial z} = 1$$

$$\frac{\partial w}{\partial x} = 0, \quad \frac{\partial w}{\partial y} = 0, \quad \frac{\partial w}{\partial z} = 1$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

49. If $x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$ show that $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = r^2 \sin \theta$

Given $x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$

$$\frac{\partial x}{\partial r} = \sin \theta \cos \phi, \quad \frac{\partial y}{\partial r} = \sin \theta \sin \phi, \quad \frac{\partial z}{\partial r} = \cos \theta$$

$$\frac{\partial x}{\partial \theta} = r \cos \theta \cos \phi, \quad \frac{\partial y}{\partial \theta} = r \cos \theta \sin \phi, \quad \frac{\partial z}{\partial \theta} = -r \sin \theta$$

$$\frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi, \quad \frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi, \quad \frac{\partial z}{\partial \phi} = 0$$

$$\begin{aligned} \therefore \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} &= \begin{vmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{vmatrix} \\ &= r^2 \sin \theta \begin{vmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{vmatrix} \\ &= r^2 \sin \theta \{ \sin^2 \theta \cos^2 \phi + \cos^2 \theta \cos^2 \phi - \sin \phi (-\sin^2 \theta \sin \phi - \cos^2 \theta \sin \phi) \} \\ &= r^2 \sin \theta \{ \cos^2 \phi (\sin^2 \theta + \cos^2 \theta) + \sin^2 \phi (\sin^2 \theta + \cos^2 \theta) \} \\ &= r^2 \sin \theta [\cos^2 \phi + \sin^2 \phi] = r^2 \sin \theta \end{aligned}$$

50. If $x = r \cos \theta$, $y = r \sin \theta$ find $J = \frac{\partial(x,y)}{\partial(r,\theta)}$ and $J' = \frac{\partial(r,\theta)}{\partial(x,y)}$. Also verify $J \cdot J' = 1$

Solution: Given $x = r \cos \theta$ and $y = r \sin \theta$

$$\Rightarrow \frac{\partial x}{\partial r} = \cos \theta; \frac{\partial x}{\partial \theta} = -r \sin \theta \quad \text{and} \quad \frac{\partial y}{\partial r} = \sin \theta; \frac{\partial y}{\partial \theta} = r \cos \theta$$

Therefore,

$$J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \quad \dots \dots \dots (1)$$

To find J' , we have to express r and θ in terms of x and y . We find that

$$x^2 + y^2 = r^2 \Rightarrow r = \sqrt{x^2 + y^2} \quad \text{and} \quad \frac{y}{x} = \tan \theta \Rightarrow \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

$$\Rightarrow \frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}; \frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}} \quad \text{and} \quad \frac{\partial \theta}{\partial x} = \frac{-y}{x^2 + y^2}; \frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2}$$

Therefore,

$$J' = \frac{\partial(r,\theta)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{\sqrt{x^2 + y^2}} & \frac{y}{\sqrt{x^2 + y^2}} \\ \frac{-y}{x^2 + y^2} & \frac{x}{x^2 + y^2} \end{vmatrix} = \frac{x^2 + y^2}{(x^2 + y^2)^{3/2}}$$

$$= \frac{1}{(x^2 + y^2)^{1/2}} = \frac{1}{r} \quad \dots \dots \dots (2)$$

From (1) and (2), we get

$$JJ' = 1$$

51. If $u = x^2 - y^2$ and $v = 2xy$ where $x = r \cos \theta$, $y = r \sin \theta$ show that $\frac{\partial(x,y)}{\partial(r,\theta)} = r$ and $\frac{\partial(u,v)}{\partial(r,\theta)} = 4r^3$.

Solution: Consider the expressions $u = x^2 - y^2$ and $v = 2xy$

$$\Rightarrow \frac{\partial u}{\partial x} = 2x; \frac{\partial u}{\partial y} = -2y \quad \text{and} \quad \frac{\partial v}{\partial x} = 2y; \frac{\partial v}{\partial y} = 2x$$

$$\text{Therefore,} \quad \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4(x^2 + y^2) \quad \dots \dots \dots (1)$$

Again consider,

$$x = r \cos \theta, \quad y = r \sin \theta \Rightarrow x^2 + y^2 = r^2$$

$$\Rightarrow \frac{\partial x}{\partial r} = \cos \theta; \frac{\partial x}{\partial \theta} = -r \sin \theta \quad \text{and} \quad \frac{\partial y}{\partial r} = \sin \theta; \frac{\partial y}{\partial \theta} = r \cos \theta$$

Therefore,

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \quad \dots \dots \dots (2)$$

Consequently, we obtain, by using (1) and (2)

$$\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(r,\theta)} = 4(x^2 + y^2) \cdot r = 4r^3 \quad (\text{using } x^2 + y^2 = r^2)$$

52. If $u = z - x$, $v = y - z$ and $w = x + y + z$ find $J = \frac{\partial(u,v,w)}{\partial(x,y,z)}$ and $J' = \frac{\partial(x,y,z)}{\partial(u,v,w)}$. Also verify $J \cdot J' = 1$

We have $u = z - x$, $v = y - z$, $w = x + y + z$

$$\frac{\partial u}{\partial x} = -1, \quad \frac{\partial u}{\partial y} = 0, \quad \frac{\partial u}{\partial z} = 1$$

$$\frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = 1, \quad \frac{\partial v}{\partial z} = -1$$

$$\frac{\partial w}{\partial x} = 1, \quad \frac{\partial w}{\partial y} = 1, \quad \frac{\partial w}{\partial z} = 1$$

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} = -1(1 + 1) + 1(-1 - 0) = -2 - 1 = -3$$

Given $u + v = y - x$, $u + v + w = z + 2y$, $u + v + w + v = 3y$

$$\Rightarrow y = \frac{1}{3}(u + 2v + w)$$

$$y - x = u + v$$

$$\Rightarrow x = y - (u + v) = \frac{1}{3}(w - 2u - v)$$

$$z - x = u \Rightarrow z = u + \frac{1}{3}(w - 2u - v) = \frac{1}{3}(3u + w - 2u - v)$$

$$z = \frac{1}{3}(w + u - v)$$

$$\frac{\partial x}{\partial u} = -\frac{2}{3}, \quad \frac{\partial x}{\partial v} = \frac{-1}{3}, \quad \frac{\partial x}{\partial w} = \frac{1}{3}$$

$$\frac{\partial y}{\partial u} = \frac{1}{3}, \quad \frac{\partial y}{\partial v} = \frac{2}{3}, \quad \frac{\partial y}{\partial w} = \frac{1}{3}$$

$$\frac{\partial z}{\partial u} = \frac{1}{3}, \quad \frac{\partial z}{\partial v} = \frac{-1}{3}, \quad \frac{\partial z}{\partial w} = \frac{1}{3}$$

$$\begin{aligned}
 J' = \frac{\partial(x, y, z)}{\partial(u, v, w)} &= \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} & \frac{-1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{-1}{3} & \frac{1}{3} \end{bmatrix} \\
 &= \frac{-2}{3} \left(\frac{2}{9} + \frac{1}{9} \right) + \frac{1}{3} \left(\frac{1}{9} - \frac{1}{9} \right) + \frac{1}{3} \left(-\frac{1}{9} - \frac{2}{9} \right) = \frac{-2}{9} - \frac{1}{9} = \frac{-1}{3} \\
 \therefore JJ' &= (-3) \left(\frac{-1}{3} \right) = 1
 \end{aligned}$$